

Esya PD360: A multi-parametric platform for establishing a novel lysosome-based biomarker for Parkinson's Disease (MJFF-022595)

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Abstract

The main goal of the project was to develop a multi-parametric platform to differentiate Parkinson's Disease (PD) from Healthy individuals (HI) by profiling i) Lysosomal pH and Ca^{2+} , ii) Autophagy index and iii) transcriptomic changes using an RNA sequencing in patient derived skin fibroblast cells. Skin fibroblasts were sourced from four cohorts: PPMI, Lubeck, Esya's banked samples and NINDS consisting of 32 HI, 83 PD (24 sporadic, 59 genetic) and 1 PD at risk samples having a SNCA mutation. Lysosomal ionic profile and Autophagy were quantified using ratiometric fluorescent probes (sensor CalipHluor and tandem sensor RFP-GFP-LC3B respectively) and were analyzed using ESYA's findosome software. Transcriptomic changes assayed using RNA-Seq analysis and relative expression levels were reported. Results from these experiments were used to build a multi-parametric PD diagnosis platform PD360.

Methods & analysis

(i) Lysosomal ionic profiling: Lysosomal Calcium and pH measurement was performed using ESYA's patented CalipHluor probe¹ following a 3-day protocol. Primary fibroblast cells were cultured and maintained using complete medium (1x DMEM, 1x Non-Essential Amino Acids, 20mM HEPES buffer pH7.4, 10% FBS and 1x Penicillin-Streptomycin) in a CO₂ incubator (37°C & 5% CO₂). On day-1, fibroblast cells were plated on a 96-well plate (5000 cells/well, in duplicate for each cell line) and allowed to grow for 16-18 hr in incubator. On the day-2 (day of imaging) cells were incubated with CalipHluor for 1.5 hr (800 nM in 80 μL) in human minimum essential medium (hMEM) at 37°C & 5% CO₂, then washed with 1x PBS (200 μL *3 times) and chased for 6 hr in complete medium to make sure the probe is trafficked to lysosome. The cells were imaged with 60x air objective using INCA6000 fluorescent microscope in green(G) (FITC channel: Ex/Em: 488 nm/520 nm) orange(O) (Cy3 channel: Ex/Em: 560 nm/580 nm), red(R) (Cy5 channel: Ex/Em: 650 nm/665 nm) and brightfield channels. To convert the G/R and O/R from the lysosomes to absolute pH and Calcium, an *in vitro* pH calibration experiment was conducted using CalipHluor coated Silica beads (0.69 μm). The beads were incubated with different pH buffers in the physiological pH range (3- 7) as well as Calcium range (0mM (low) and 20 mM(high)) and imaged

along with cells using the same settings¹. Absolute pH and Calcium values per lysosome were calculated as described in the analysis section.

Analysis: Findosome software was used to find out the background subtracted intensities for beads in G, O and R channels; and converting them to G/R and O/R values in each buffer condition. The G/R and O/R from the lysosomes were also calculated in a similar way. The ratio of G/R for the beads was plotted against individual known pH's to generate pH calibration curve and fitted to a sigmoidal plot. The midpoint of curve as pKa, slope of the curve, high G/R (at pH 7) and low G/R (at pH 3) were calculated. The G/R from lysosomes were converted to absolute pH using the equation, $pH = pKa + \text{slope} * \ln(\text{lowG/R} - G/R) / (G/R - \text{highG/R})$.

Absolute Calcium levels the lysosomes were calculated using equation, $[Ca] = Kd * (O/R - \text{low O/R}) / (\text{high O/R} - O/R)$. Where O/R is calculated for each lysosome while low O/R and high O/R were calculated from beads at pH 4.5, with 0 mM or 20 mM free Calcium respectively.

Kd, for calcium binding was calculated using the equation $= 1.03 + 5.14 * 10^{12} * e^{(-pH/0.189)} + 3.108 * 10^6 * e^{(-pH/0.412)}$.

For pH, Calcium and pH/Ca²⁺ mean of two wells were reported.

(ii) Autophagy: Autophagy was assayed using Tandem Sensor RFP-GFP-LC3B (Cat No P36239, Thermo Fisher)² using a 3-day protocol. Primary fibroblast cells were cultured and maintained using complete medium (1x DMEM, 1x Non-Essential Amino Acids, 20mM HEPES buffer pH7.4, 10% FBS and 1x Penicillin-Streptomycin) in a CO₂ incubator (37°C & 5% CO₂). On day-1, primary fibroblast cells were plated in a 96-well plate (2500 cells/ well, 4 wells per cell lines; 2 wells for autophagy induction and 2 for DMSO control) allowed to grow for 16-18 hr in incubator. On day-2, the cells were infected with PREMO Autophagy Tandem sensor (3 µL/well in complete medium, 3*10⁵ particles per ~4000 cells), for 24 hr. On the day of imaging (day-3) cells were induced for Autophagy using 1µM Torin-1 (Abcam, ab218606) for 6h or corresponding DMSO control (2 wells for each condition) in complete medium. Fluorescent imaging was done using Olympus IX83 inverted microscope, using 100x oil objective in GFP (FITC, Ex/Em: 488 nm/520 nm), RFP (Cy3, Ex/Em: 560 nm/580 nm) and brightfield channels. ESYA's analysis software, findosome was used for the analysis.

Analysis: The ratio of intensities of GFP/RFP (G/R) per autophagosome, number of autophagosomes/cells and area of autophagosomes was calculated using ESYA's findosome software for each well. Median GFP/RFP per well were calculated and mean of two wells were reported for induced (torin-1 treated) or uninduced (DMSO treated) cases. For Autophagy index, the mean G/R of Induced/ Uninduced was reported. Similarly for number of autolysosome and area of autolysosomes, mean ratio of induced/uninduced number of autophagosomes/cell or Area of autophagosomes were reported respectively. For particle number, mean of number of yellow particles/total particles per cell (With G/R >0.1 of individual autophagosome to total) was

calculated from 2 wells for both induced and uninduced cases. The mean ratio of induced particle number/cell to uninduced was reported as particle autology.

(iii) Gene expression levels: Gene expression levels were assayed using RNA-sequencing. It contains the following steps,

- a) **RNA Extraction:** Total RNA was extracted from frozen cell pellets using Qiagen RNeasy Mini kit following manufacturer's instructions (Qiagen, Hilden, Germany). RNA samples were quantified using Qubit 4.0 Fluorometer (Life Technologies, Carlsbad, CA, USA) and RNA integrity was checked with RNA Kit on Agilent 5300 Fragment Analyzer (Agilent Technologies, Palo Alto, CA, USA). ERCC RNA Spike-In Mix kit (cat. 4456740) from Thermo Fisher Scientific/Invitrogen, was added to normalized total RNA prior to library preparation following manufacturer's protocol.
- b) **Library Preparation and NovaSeq Sequencing:** RNA sequencing libraries were prepared using the NEBNext Ultra II RNA Library Prep Kit for Illumina following manufacturer's instructions (NEB, Ipswich, MA, USA). Briefly, mRNAs were first enriched with Oligo(dT) beads. Enriched mRNAs were fragmented for 15 minutes at 94 °C. First strand and second strand cDNAs were subsequently synthesized. cDNA fragments were end repaired and adenylated at 3'ends, and universal adapters were ligated to cDNA fragments, followed by index addition and library enrichment by limited-cycle PCR. Sequencing libraries were validated using NGS Kit on the Agilent 5300 Fragment Analyzer (Agilent Technologies, Palo Alto, CA, USA), and quantified by using Qubit 4.0 Fluorometer (Invitrogen, Carlsbad, CA). The sequencing libraries were multiplexed and loaded on the flowcell on the Illumina NovaSeq 6000 instrument according to manufacturer's instructions. The samples were sequenced using a 2x150 Pair-End (PE) configuration v1.5. Image analysis and base calling were conducted by the NovaSeq Control Software v1.7 on the NovaSeq instrument. Raw sequence data (.bcl files) generated from Illumina NovaSeq was converted into fastq files and de-multiplexed using Illumina bcl2fastq program version 2.20. One mismatch was allowed for index sequence identification.

Analysis: After investigating the quality of the raw data, sequence reads were trimmed to remove possible adapter sequences and nucleotides with poor quality using Trimmomatic v.0.36. The trimmed reads were mapped to the Homo sapiens reference genome available on ENSEMBL using the STAR aligner v.2.5.2b. The STAR aligner is a splice aligner that detects splice junctions and incorporates them to help align the entire read sequences. BAM files were generated because of this step. Unique gene hit counts were calculated by using feature Counts from the Subread package v.1.5.2. Only unique reads that fell within exon regions were counted.

After extraction of gene hit counts, the gene hit counts table was used for downstream differential expression analysis. Using DESeq2, a comparison of gene expression between the groups of samples were performed. The Wald test was used to generate p-values and Log2 fold changes. Genes with p-values < 0.05 and variable log2 fold changes (due to large nature of different cohorts and mutations) were called as differentially expressed, genes for each comparison. A gene ontology analysis was performed on the statistically significant set of genes by implementing the software GeneSCF. The goa_human GO list was used to cluster the set of genes based on their biological process and determine their statistical significance. A PCA analysis was performed using the "plotPCA" function within the DESeq2 R package. The plot shows the samples in a 2D plane spanned by their first two principal components. The top 500 genes, selected by highest row variance, were used to generate the plot.

References

1. Narayanaswamy, N., Chakraborty, K., Saminathan, A. *et al.* A pH-correctable, DNA-based fluorescent reporter for organellar calcium. *Nat Methods* **16**, 95–102 (2019).
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2. [Premo Autophagy Assays with Sensor LC3B, p62, or Tandem Sensor RFP-GFP-LC3B \(thermofisher.com\)](https://www.thermofisher.com)

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